

FactorySwarm

Multimodal specialist agents for fast manufacturing inspection on Cerebras.

Decision support only - human verification required. Visual inspection cannot establish electrical functionality.

Inputs

Load Sample Case

Golden-reference image

reference.jpg221.3KB

+

Inspection image

inspection.jpg217.2KB

+

Asset type

Inspection stage

Reported symptom

Dataset annotation mask

Upload

200MB per file • JPG, PNG

> Optional corresponding ROI crops

Images

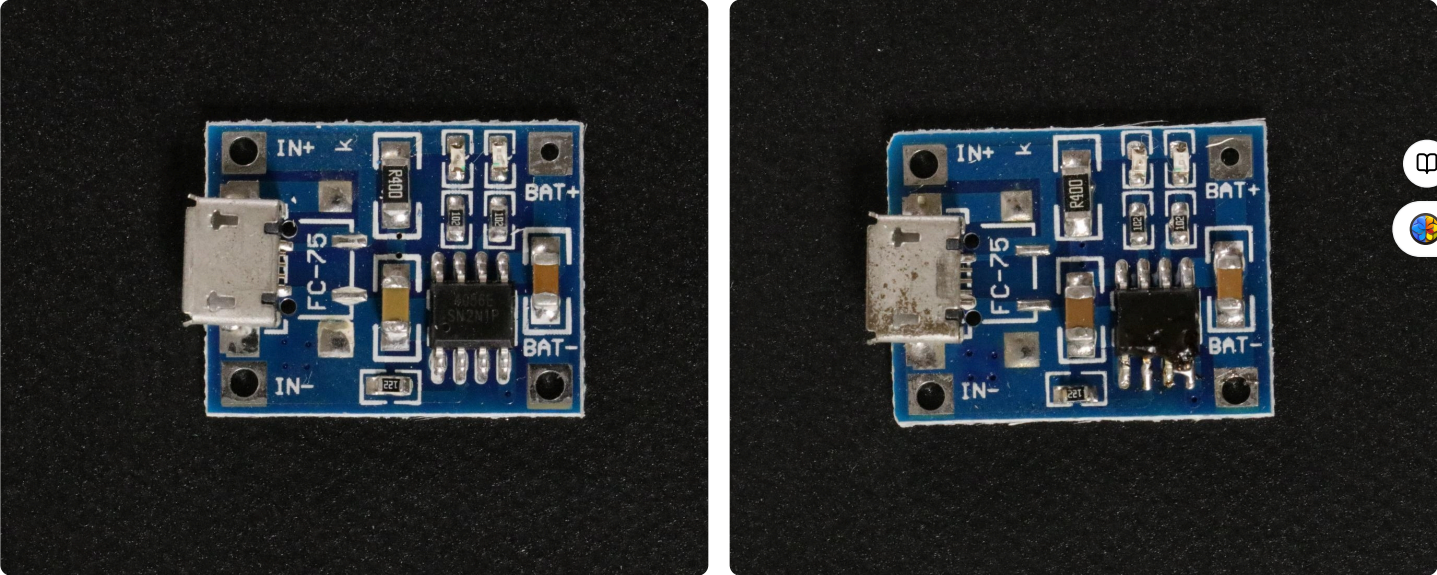


Image 1: Golden Reference

Image 2: Inspected Product

Agent Execution

Visual Difference Inspector

COMPLETED

Component and Layout Inspector

COMPLETED

Quality and Risk Assessor

COMPLETED

Factory Action Planner

COMPLETED

Run Inspection

Final Consensus

MANUAL REVIEW

Confidence 70%

There is a clear and unambiguous difference in the package style and markings of the central IC between the golden reference and the inspected product. While this is a significant discrepancy, it cannot be definitively classified as a defect (reject) without the BOM, as it could be a valid engineering revision or an approved alternative part. Therefore, manual review is required to arbitrate against the BOM. Policy safeguards applied: Confidence capped at 0.75: full-image comparison without corresponding ROI crops. Confidence capped at 0.70: ambiguous component or package comparison.

Confirmed Observations

- The central IC in the inspected product (Image 2) is a black, multi-pin surface mount package (SOIC style) with visible leads on the sides.
- The central IC in the golden reference (Image 1) is a smaller, rectangular package with a light-colored top surface and markings '4056E SN2N1P'.

Hypotheses

- The component in Image 2 may be an incorrect part populated during assembly.
- The component in Image 2 may be an approved alternative part or a hardware revision not reflected in the golden reference.

Contradictions

- None reported.

Unsupported Claims Removed

- Claims of 'through-hole' package (contradicted by visual evidence of surface mount leads).

Immediate Actions

- Quarantine the inspected unit to prevent further processing.

Follow-Up Actions

- Verify the Bill of Materials (BOM) and engineering change orders (ECO) to determine if the SOIC package is an approved substitute for the reference component.

Missing Evidence

- Current Bill of Materials (BOM)
- Approved Component Substitution List

Confidence-Cap Explanations

- Prefer manual_review for ambiguous package or component differences when BOM context is missing.
- Confidence capped at 0.75: full-image comparison without corresponding ROI crops.
- Confidence capped at 0.70: ambiguous component or package comparison.

Human Review

Required



Specialist Reports

Visual Difference Inspector

Comparison between the golden reference and the inspected product reveals a significant difference in the central integrated circuit (IC) component.

Decision: MANUAL REVIEW

Confidence: 95%

Findings

- Different IC package/appearance (center of PCB): The central IC in Image 2 is a black, multi-pin surface mount package with visible leads on the sides, whereas the IC in Image 1 is a smaller, rectangular package with a light-colored top surface and markings 'SN2N1P'.

Limitations

- Lack of BOM context to determine if the IC variation is an approved revision.

Component and Layout Inspector

Comparison between the golden reference and the inspected product reveals a significant component discrepancy in the central IC area.

Decision: MANUAL REVIEW

Confidence: 95%

Findings

- Component mismatch/replacement in central IC position (central IC area): The golden reference (Image 1) shows a surface-mount IC with visible markings '4056E SN2N1P'. The inspected product (Image 2) features a different package style (through-hole or leaded package) with a black epoxy body and no visible markings, occupying the same functional region.

Limitations

- Unable to determine if the component is functionally equivalent without a BOM.

Quality and Risk Assessor

Comparison between the golden reference and the inspected product reveals a significant component mismatch. The inspected product features a different integrated circuit (IC) package and marking compared to the reference.

Decision: REJECT

Confidence: 95%

Findings

- Component mismatch at IC location (Center-right IC area): The golden reference (Image 1) shows a surface-mount IC with clear markings '4056E SN2N1P'. The inspected product (Image 2) features a different IC package with a dark, glossy top and different lead configuration, lacking the same visible markings.

Limitations

- Lighting differences between images
- Lack of BOM to confirm if the change is an approved revision

Factory Action Planner

The inspected product shows a significant component discrepancy compared to the golden reference. A large 8-pin IC is present in the inspected product where the reference image shows a smaller, differently marked component.

Decision: MANUAL REVIEW

Confidence: 100%

Findings

- Component mismatch in central IC area (center-right): Image 2 shows a black 8-pin SOIC package, whereas Image 1 shows a smaller component with markings '4056E SN2N1P'.

Limitations

- Visual inspection cannot determine if the substituted component is functionally equivalent or if it causes electrical shorts.

Performance

Parallel Stage	Verifier	Total	Estimated Speedup
1.42s	1.11s	2.73s	3.41x

Sequential latency and speedup are estimates from per-agent timings.

Agent	Latency seconds
Visual Difference Inspector	1.42
Component and Layout Inspector	1.06
Quality and Risk Assessor	1.09
Factory Action Planner	1.27

Estimated sequential specialist latency: 4.83s

